

AQuery, KDB-X, and ReproZip course material

Prepared for Advanced Database Systems, Fall 2026.

AQuery runs in Docker on Mac or Linux. ReproZip only works on x86 Linux. Mac students need the course compute server for the ReproZip project.

These notes replace the older AQuery and ReproZip material.

Files to share

- `AQUERY_KDBX_LECTURE_NOTES.md` explains AQuery, KDB-X, q, licensing, Docker setup, and the compile-and-run workflow.
- `REPROZIP_LECTURE_NOTES.md` explains the ReproZip workflow and the commands students need for a project submission.
- `REPROZIP_TUTORIAL.md` is a worked example that traces, packs, inspects, unpacks, and reruns a small Python program.

Platform support

Workflow	Mac M-series	Mac M-series Docker	x86 Linux
AQuery and KDB-X	Works	Works	Works
ReproZip tracing	Untested	Failed (trace)	Works

The AQuery image ran successfully with Mac M-series Docker on an M1 Mac. Native KDB-X and AQuery also worked on that Mac, but the course instructions use Docker so every student follows the same workflow.

ReproZip requires Linux `ptrace`. With Mac M-series Docker on the tested M1 Mac, its AMD64 container reported invalid emulated syscalls and crashed during tracing even with `SYS_PTRACE`. Students must use an x86 Linux host for ReproZip.

Students using CIMS should connect through `access.cims.nyu.edu`, then move to the compute server assigned for the course. They should not compile or run coursework on the access gateway. Course staff must confirm that the assigned compute server provides Docker and permits `ptrace`.

Installation links for the course website

- Docker Desktop: <https://docs.docker.com/desktop/>
- KDB-X and q: https://code.kx.com/kdb-x/get_started/kdb-x-install.html
- Course repository: <https://github.com/gpu004/advance-database>
- ReproZip: <https://github.com/VIDA-NYU/reprozip#quickstart>
- ReproUnzip: <https://reprozip.readthedocs.io/en/latest/unpacking.html>
- NYU CIMS access servers: <https://cims.nyu.edu/dynamic/systems/resources/accessservers/>

Validation record

The Docker controls passed in an x86 Linux environment running x86-64 Debian 12 on August 30, 2026.

- AQuery generated an 8,002-byte q program from the three-row sales example.
- KDB-X ran the generated program and returned `banana = 7` and `coffee = 12` for the `amount > 5` filter.

- ReproZip 1.3.2 traced and packed a Python command.
- ReproUnzip 1.3.2 inspected, unpacked, and reran the command.
- The traced and reproduced outputs both contained `REPROZIP ON NYU LINUX`.

The AQuery Docker workflow also passed with Mac M-series Docker on an M1 Mac with Docker Desktop 4.87. The ReproZip trace failed there and is intentionally excluded from the Mac M-series Docker instructions.

The remaining host-specific check is whether the assigned NYU compute server provides Docker and permits ReproZip to use `ptrace`.